

第十四讲：深度学习与图像分割



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知识回顾: 常见视觉任务

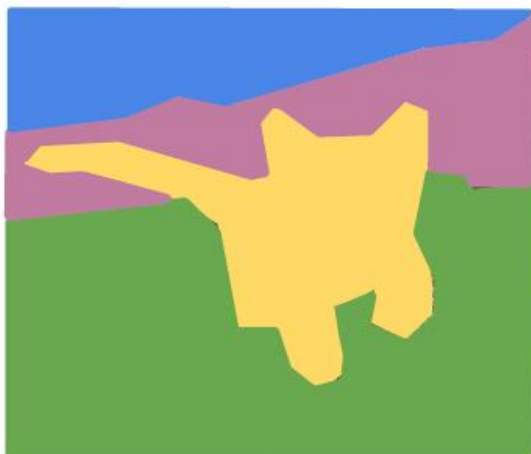
Classification
分类



CAT

No spatial extent

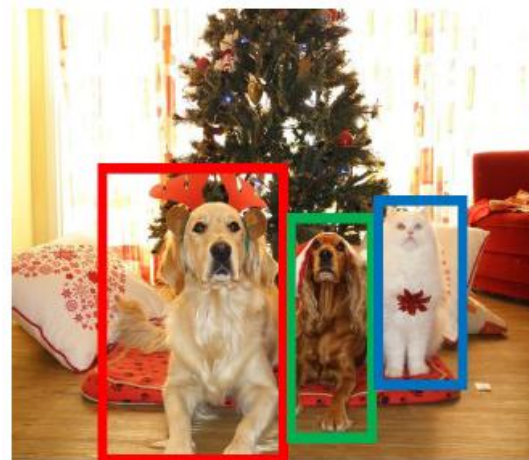
Semantic segmentation
场景分割



CAT, GRASS,
TREE, SKY

No objects, just pixels

Object detection
物体检测



DOG, DOG, CAT

Instance segmentation
实例分割

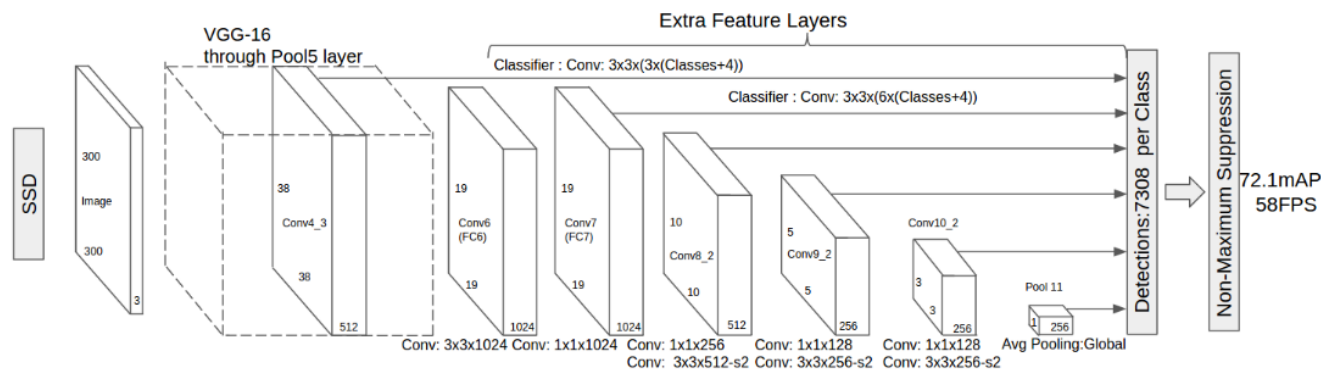
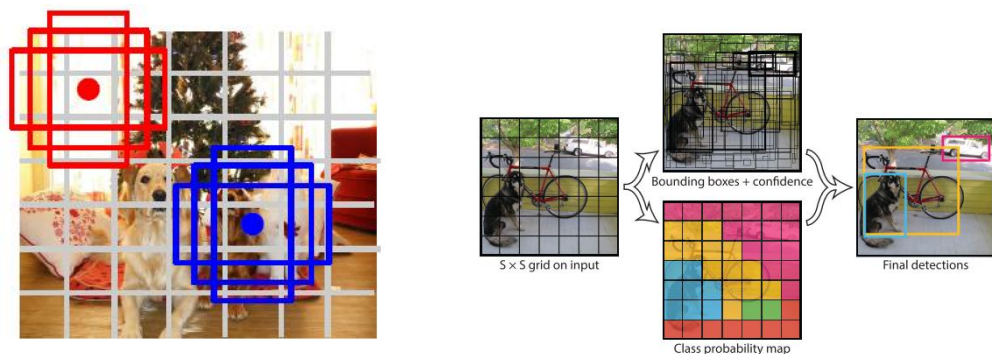
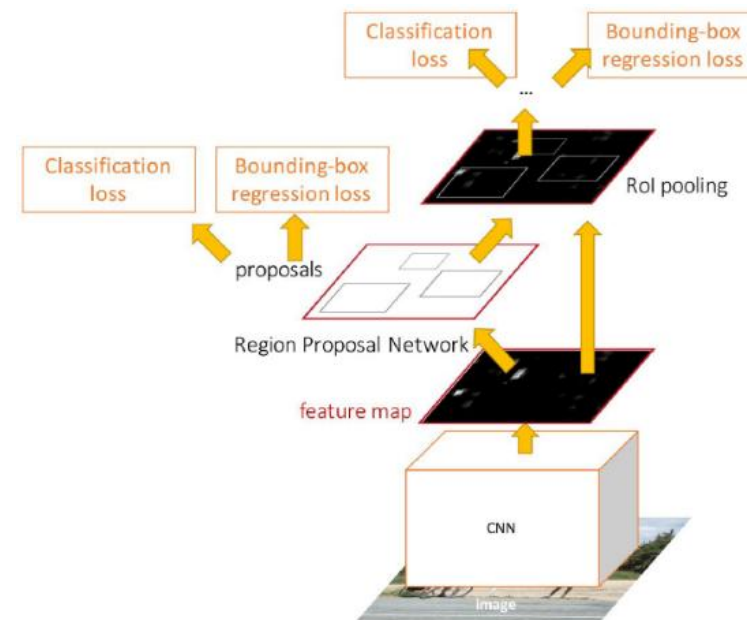
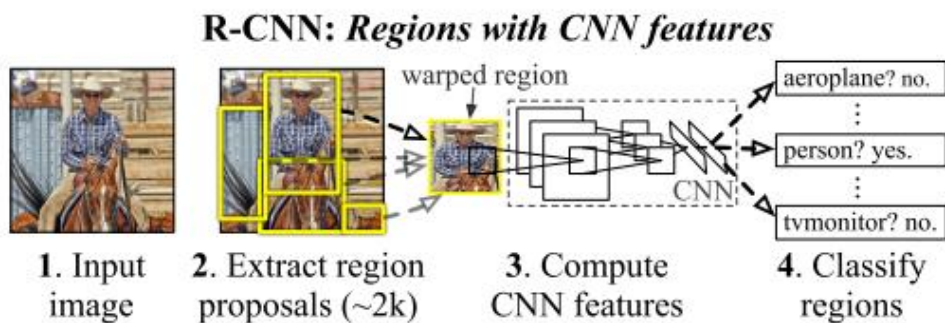


DOG, DOG, CAT

Multiple Object

知识回顾：CNN与目标检测

- R-CNN, Fast R-CNN, Faster R-CNN
- SSD, YOLO



Content



1, Image Segmentation

- FCN, U-Net, HRNet, Deeplab

2, Instance segmentation

- Mask-RCNN

3, Transfer learning

Image Segmentation 图像分割

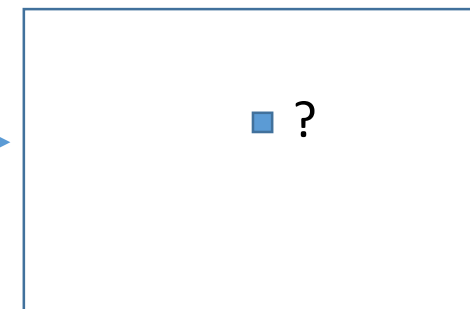


Semantic Segmentation 预测各个像素的类别



GRASS, CAT,

TREE, SKY, ...



At test time, classify each pixel of a new image.

Paired training data: for each training image,
each pixel is labeled with a semantic category

Image Segmentation 图像分割

Semantic Segmentation 基于局部特征预测像素类别



Classify center pixel with CNN



COW



grass

Problem: Very inefficient! Not reusing shared features between overlapping patches

Image Segmentation 图像分割

From image classification to segmentation

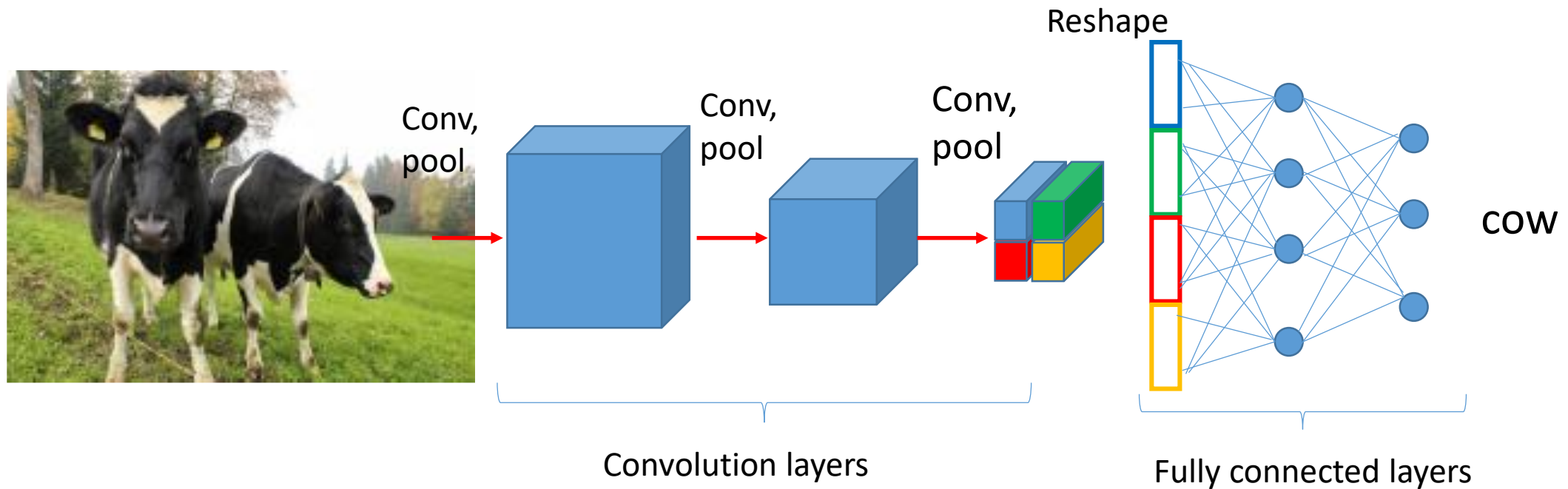
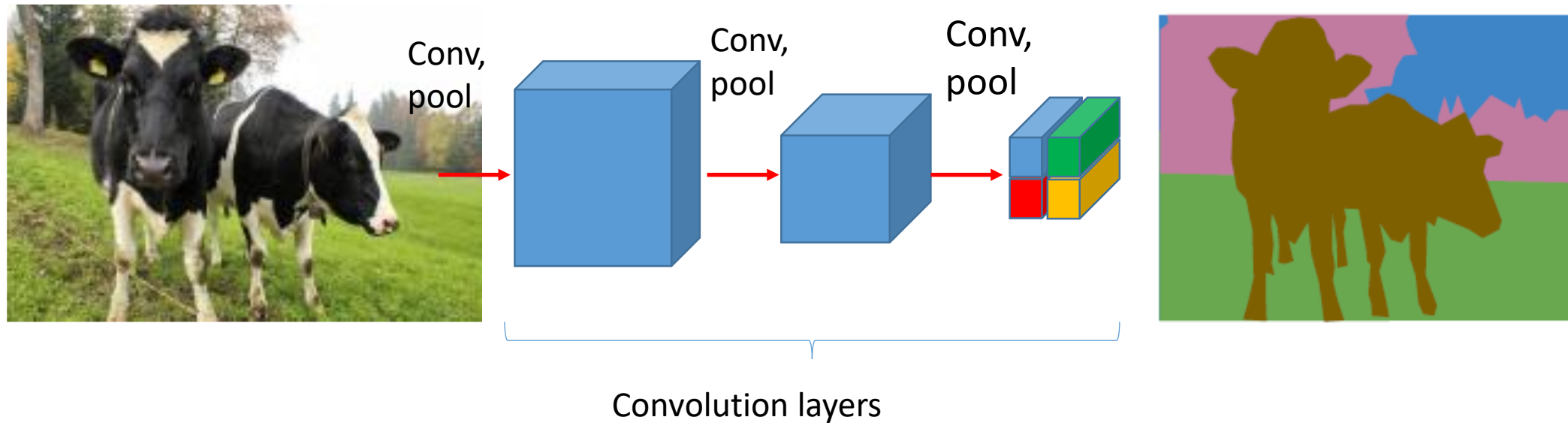


Image Segmentation 图像分割

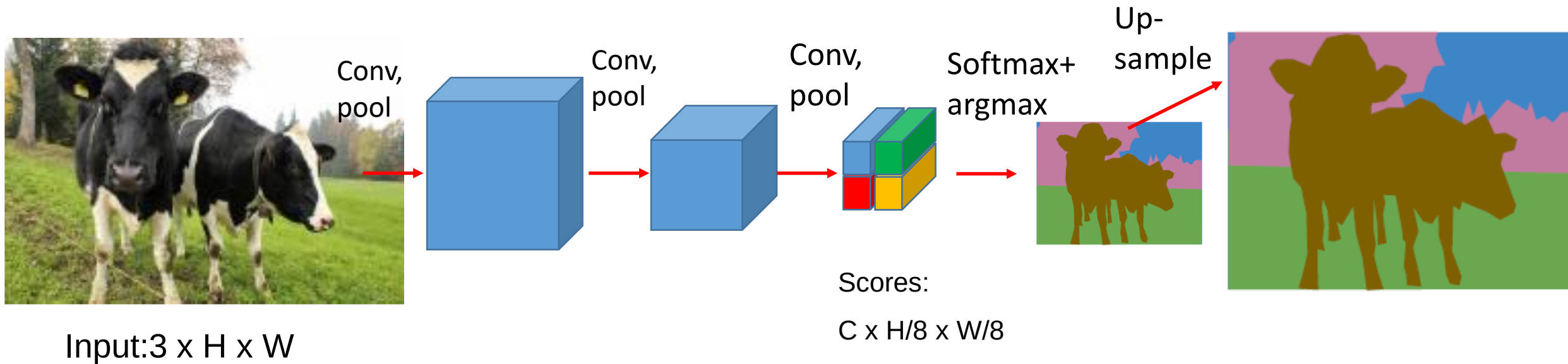
From image classification to segmentation



An intuitive idea: encode the entire image with conv net, and do semantic segmentation on top.

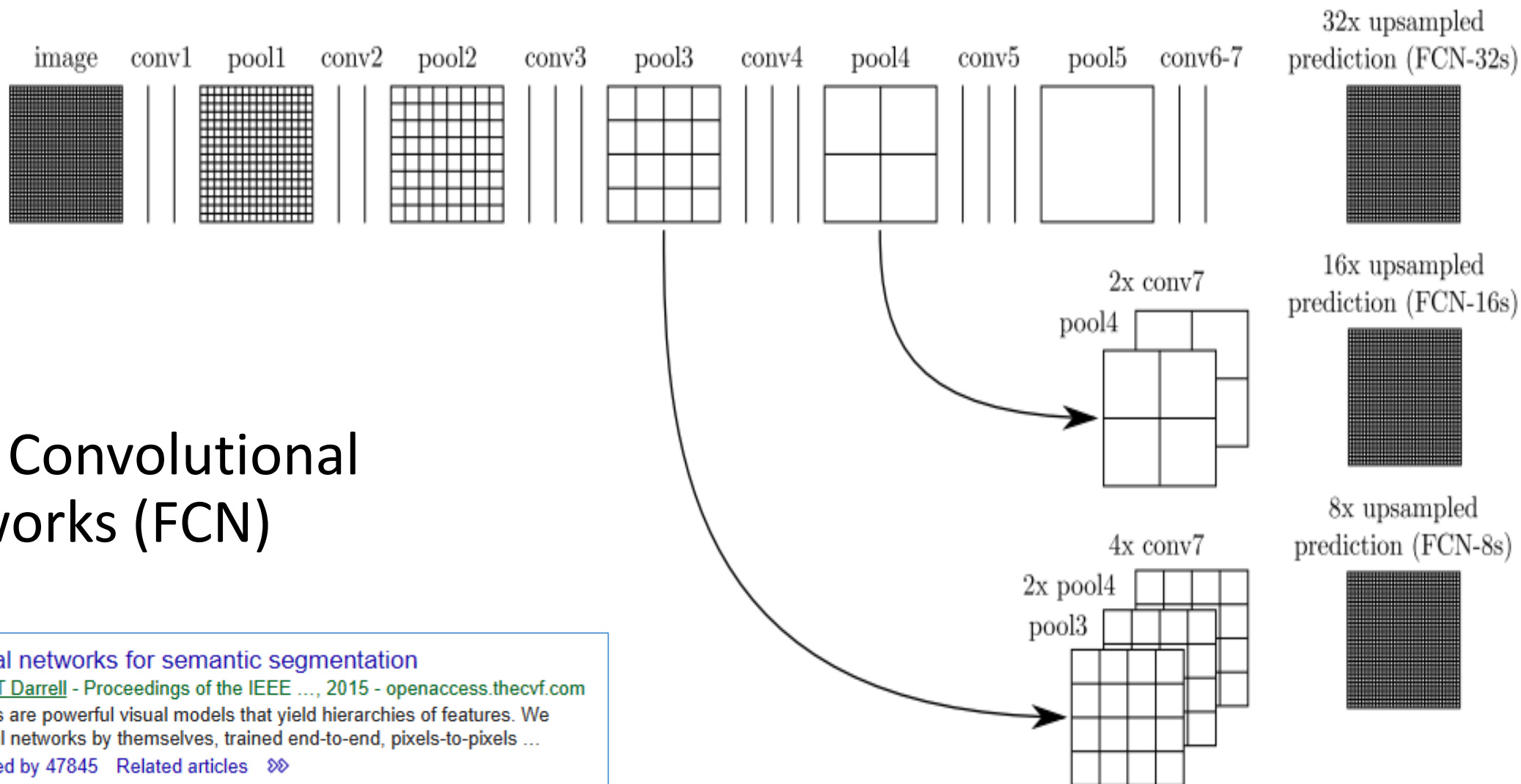
Image Segmentation 图像分割

From image classification to segmentation



An intuitive idea: encode the entire image with conv net, and do semantic segmentation on top.

Image Segmentation 图像分割



Fully Convolutional Networks (FCN)

Fully convolutional networks for semantic segmentation

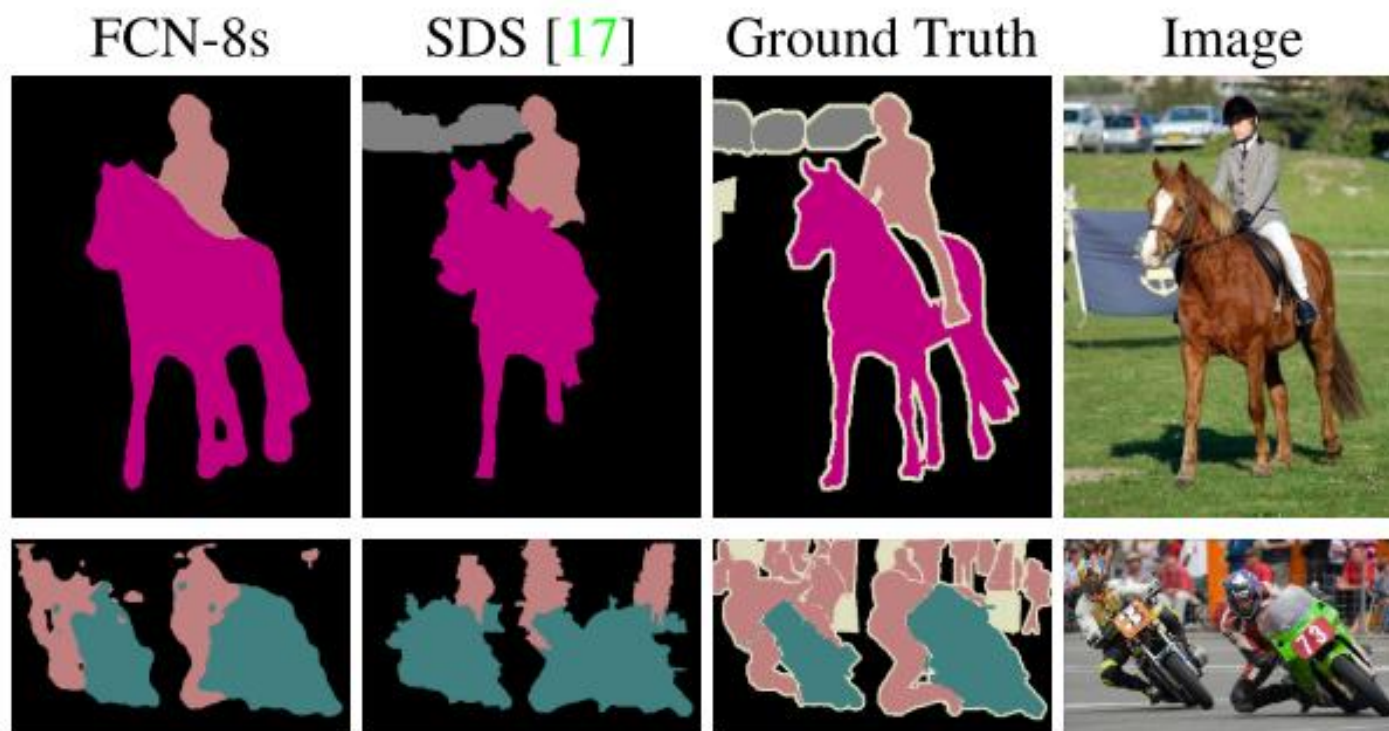
J Long, E Shelhamer, T Darrell - Proceedings of the IEEE ..., 2015 - openaccess.thecvf.com

Convolutional networks are powerful visual models that yield hierarchies of features. We show that convolutional networks by themselves, trained end-to-end, pixels-to-pixels ...

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Image Segmentation 图像分割

- Results of FCN



局限： 分割结果不够精细，如何得到更高分辨率的分割？

Image Segmentation

- Deconvolution for segmentation 反卷积
 - Design network as a bunch of convolutional layers, with **downsampling** and **upsampling** inside the network!

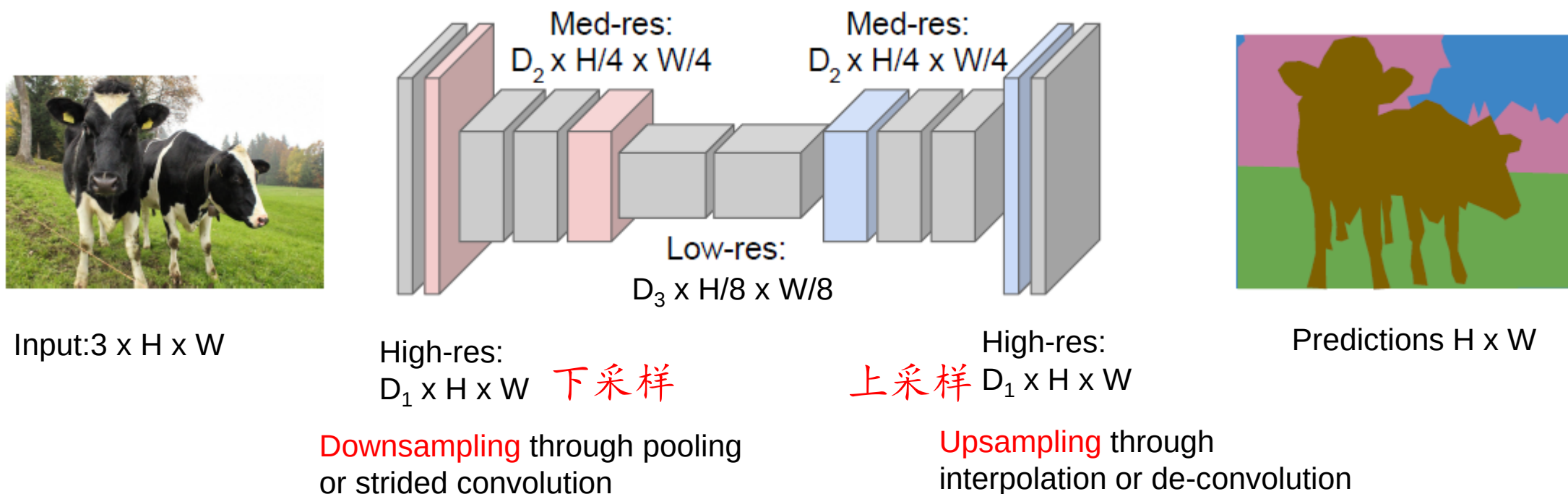


Image Segmentation

- In-Network upsampling

Nearest Neighbor

1	2
3	4



1	1	2	2
1	1	2	2
3	3	4	4
3	3	4	4

Input: 2 x 2

Output: 4 x 4

“Bed of Nails”

1	2
3	4



1	0	2	0
0	0	0	0
3	0	4	0
0	0	0	0

Input: 2 x 2

Output: 4 x 4

Image Segmentation

- In-Network upsampling: “Max Unpooling”

Max Pooling

Remember which element was max!

1	2	6	3
3	5	2	1
1	2	2	1
7	3	4	8

5	6
7	8

Rest of the network

1	2
3	4

Max Unpooling

Use positions from pooling layer

0	0	2	0
0	1	0	0
0	0	0	0
3	0	0	4

Corresponding pairs of downsampling and upsampling layers

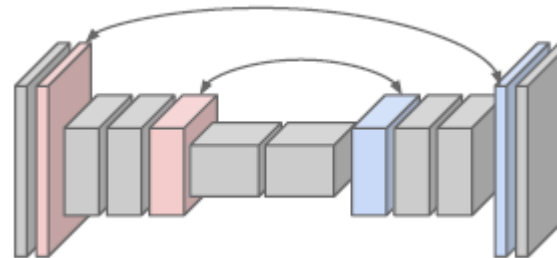


Image Segmentation

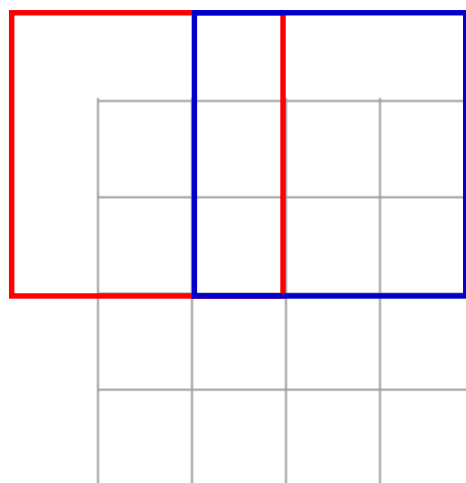


Deconvolution (transpose convolution) 反卷积

- Used to upsample the feature map

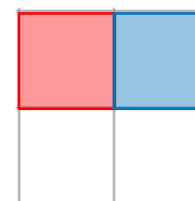
Recall: Normal 3 x 3 convolution, stride 2 pad 1

步长为2的卷积
可实现下采样
(可学习)



Input 4x4

Weighted sum
between filter
and input



Output 2x2

Filter moves 2 pixels in
the input for every one
pixel in the output

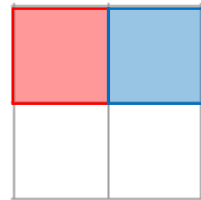
We can interpret strided
convolution as “**learnable
downsampling**”.

Image Segmentation

Deconvolution (transpose convolution) 反卷积

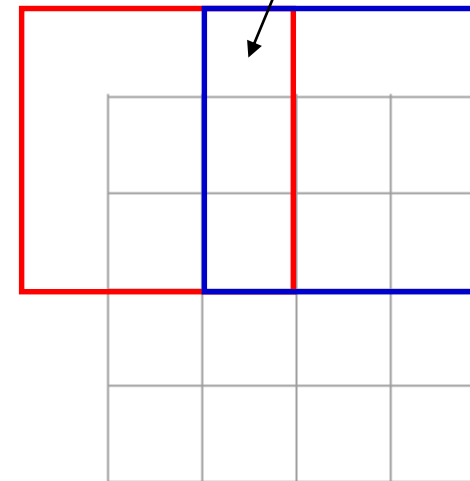
- Used to upsample the feature map

3 x 3 **transpose** convolution
stride 2 pad 1



Input 2x2

Input gives
weight for
filter



Output 4x4

Sum where output overlaps

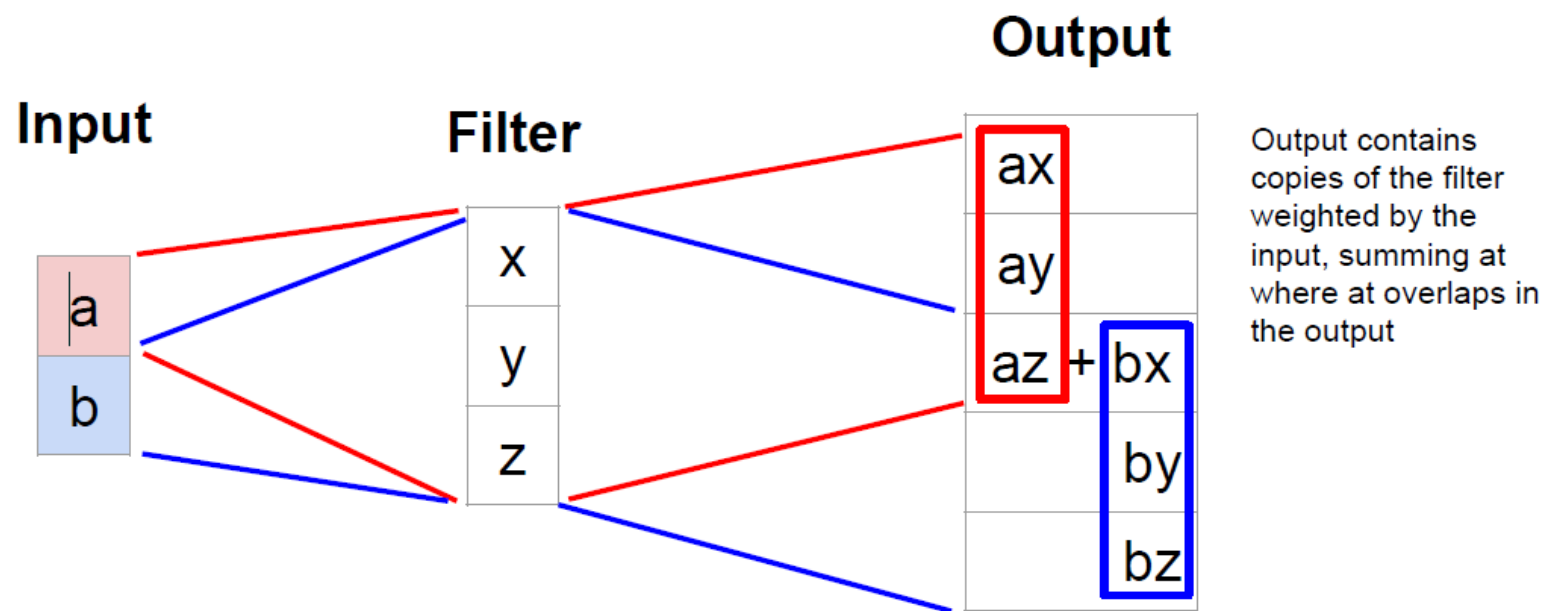
Filter moves 2 pixels
in the output for every
one pixel in the input

Stride gives ratio
between movement in
output and input

Image Segmentation

Deconvolution (transpose convolution) 反卷积

- Used to upsample the feature map



Learnable Upsampling: 1D Example

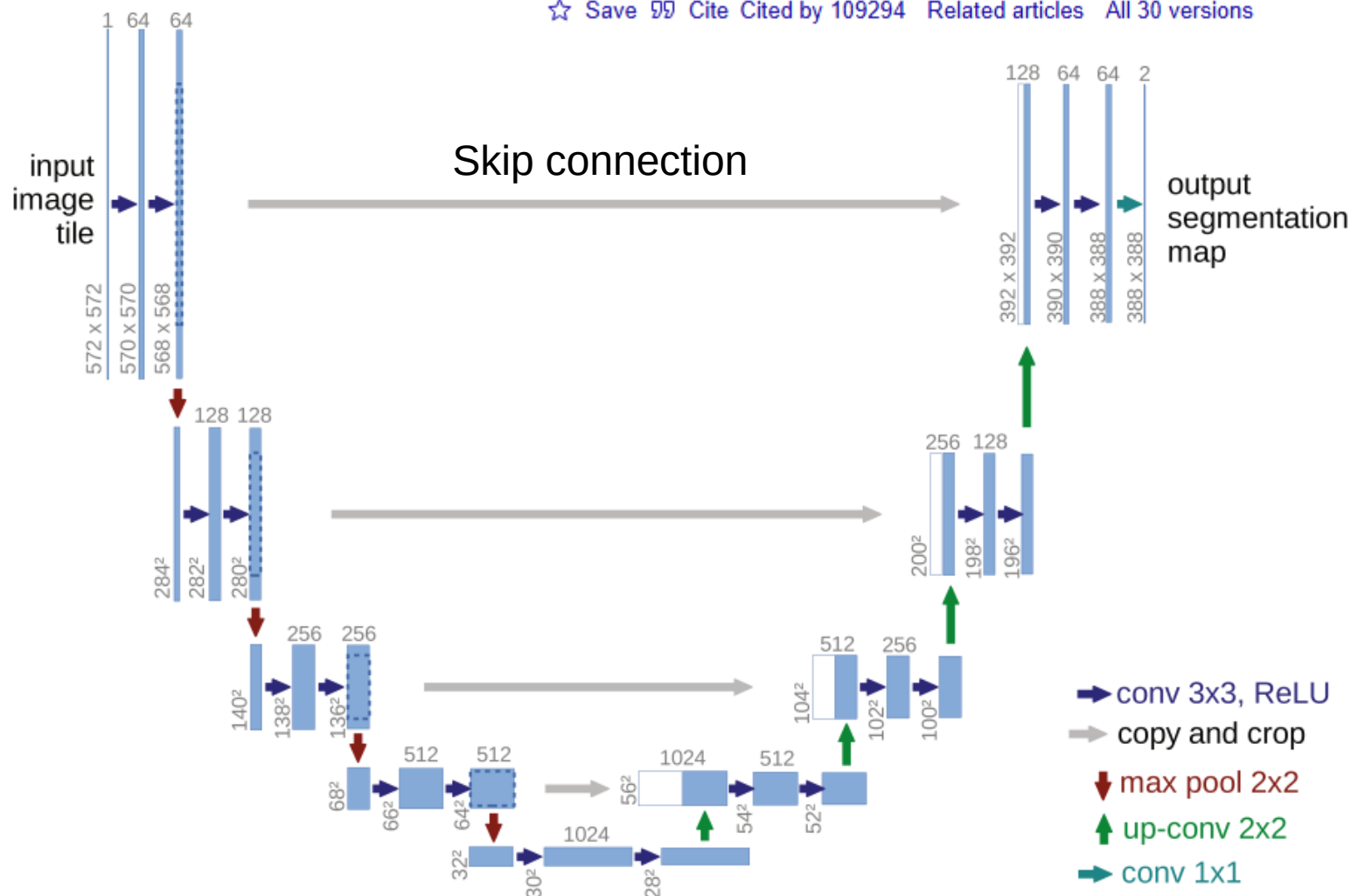
Image Segmentation

U-Net

编码器 Encoding path
(left) uses convolution and pooling to extract features

解码器 Decoding path
(right) uses deconvolution to recover the resolution

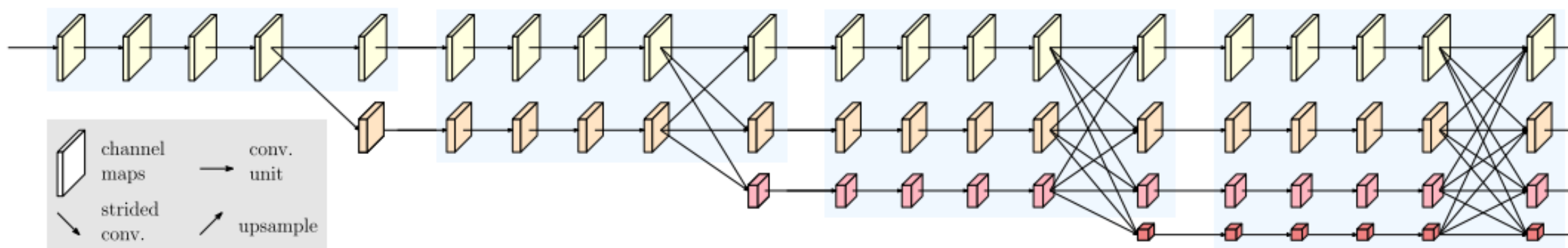
跳跃连接 skip connection
improves gradient backpropagation



U-net: Convolutional networks for biomedical image segmentation
O Ronneberger, P Fischer, T Brox - ... , Munich, Germany, October 5-9, 2015 ... , 2015 - Springer
There is large consent that successful training of deep networks requires many thousand annotated training samples. In this paper, we present a network and training strategy that relies on the strong use of data augmentation to use the available annotated samples more efficiently. The architecture consists of a contracting path to capture context and a symmetric expanding path that enables precise localization. We show that such a network can be trained end-to-end from very few images and outperforms the prior best method (a sliding ...
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Image Segmentation

HRNet



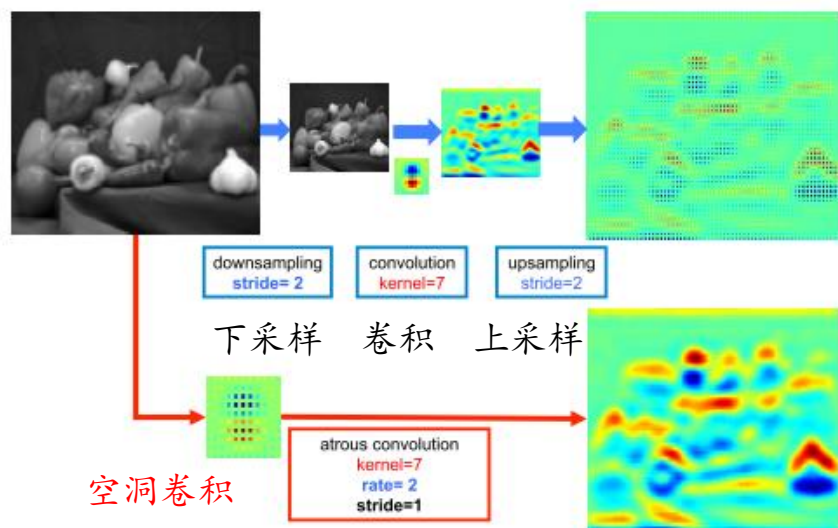
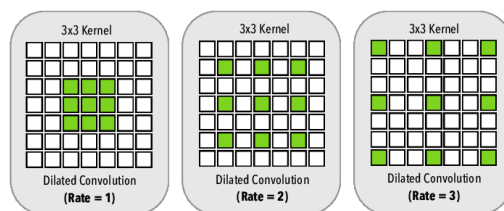
- 始终保持特征图的高分辨率，以防空间细节丢失
- 在不同尺度上进行保持分辨率的多层卷积，并融合多尺度特征



Image Segmentation

DeepLab

- 使用空洞卷积得到高分辨率、大感受野的特征



- 多个并行分支（使用不同的空洞率）得到不同尺度的特征

Atrous Spatial Pyramid Pooling (ASPP)

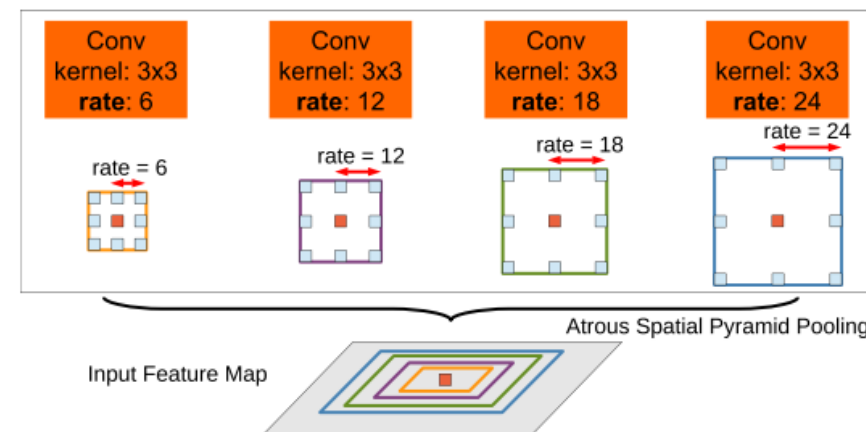
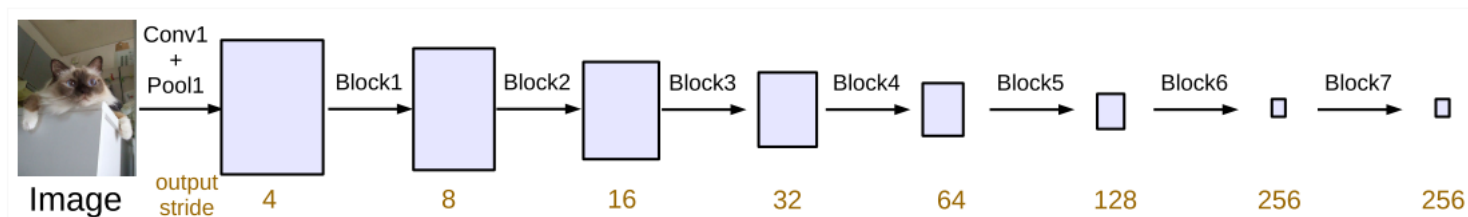


Image Segmentation

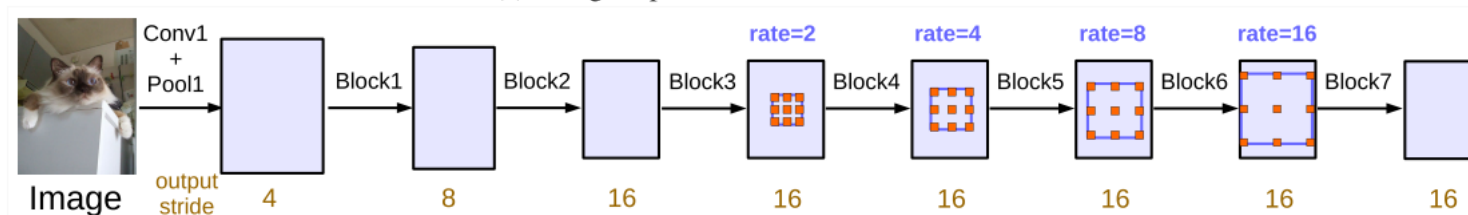
DeepLab-v3

- 多次下采样造成分辨率丢失



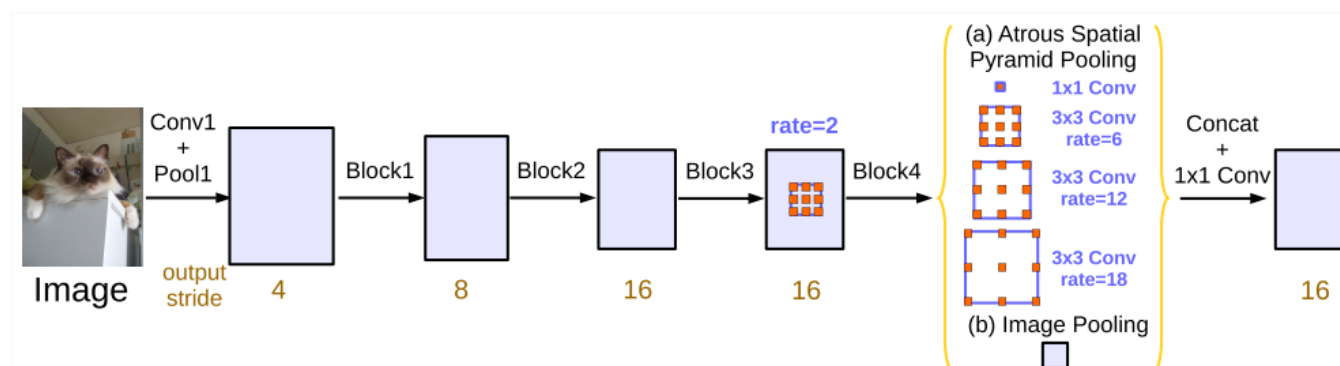
(a) Going deeper without atrous convolution.

- 使用空洞卷积



(b) Going deeper with atrous convolution. Atrous convolution with $rate > 1$ is applied after block3 when $output_stride = 16$.

- 不同空洞率的空洞卷积



Content



1, Image Segmentation

- FCN, U-Net, HRNet, Deeplab

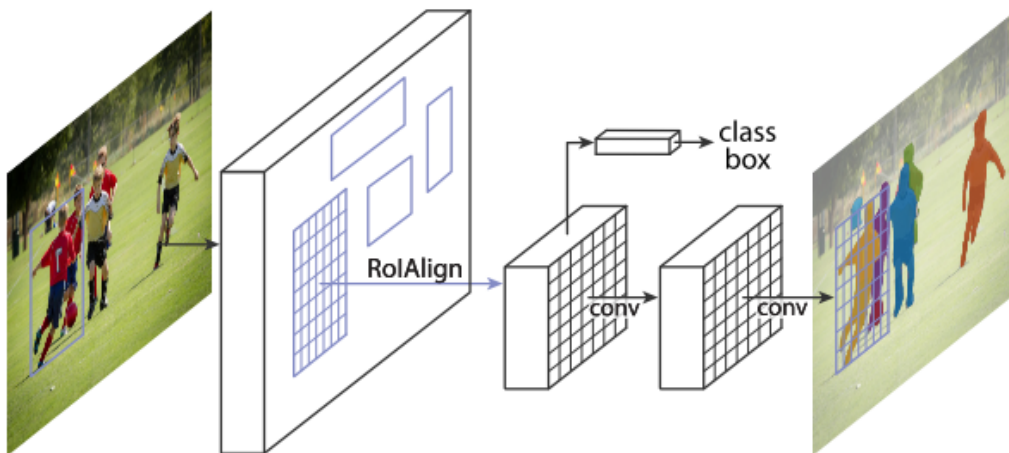
2, Instance segmentation

- Mask-RCNN

3, Transfer learning

Instance Segmentation 实例分割

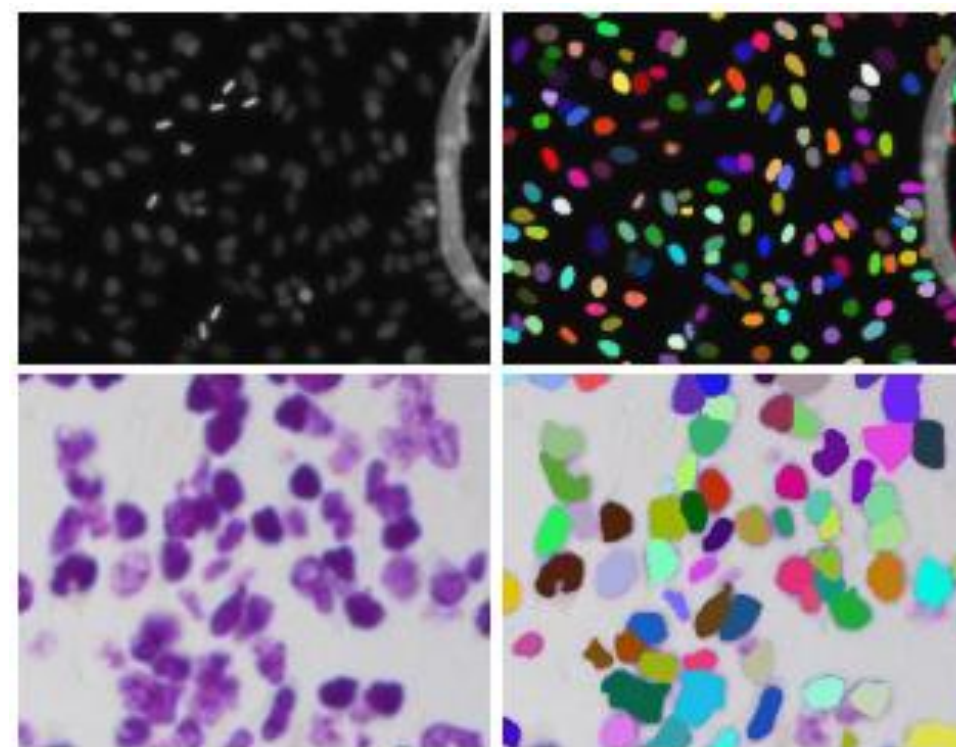
From Faster R-CNN to Mask R-CNN



- 对于同一个类别的物体，区分出不同的个体

Mask r-cnn

[K He, G Gkioxari, P Dollár...](#) - Proceedings of the IEEE ..., 2017 - openaccess.thecvf.com
... mask, with minimal modification Mask R-CNN can be applied to detect instance-specific poses.
Without tricks, Mask R-CNN ... masks on the top 100 detection boxes, Mask R-CNN adds a ...
☆ Save 77 Cite Cited by 34112 Related articles All 22 versions 77



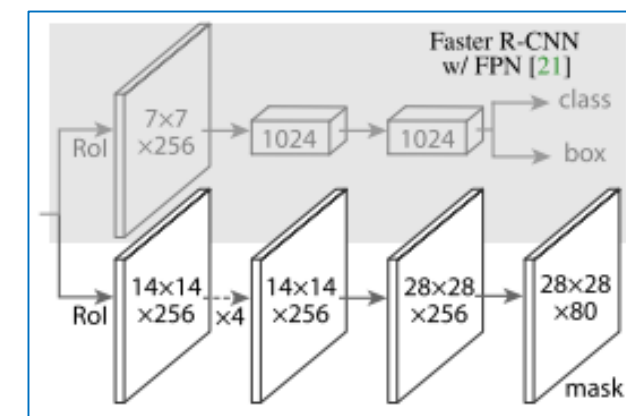
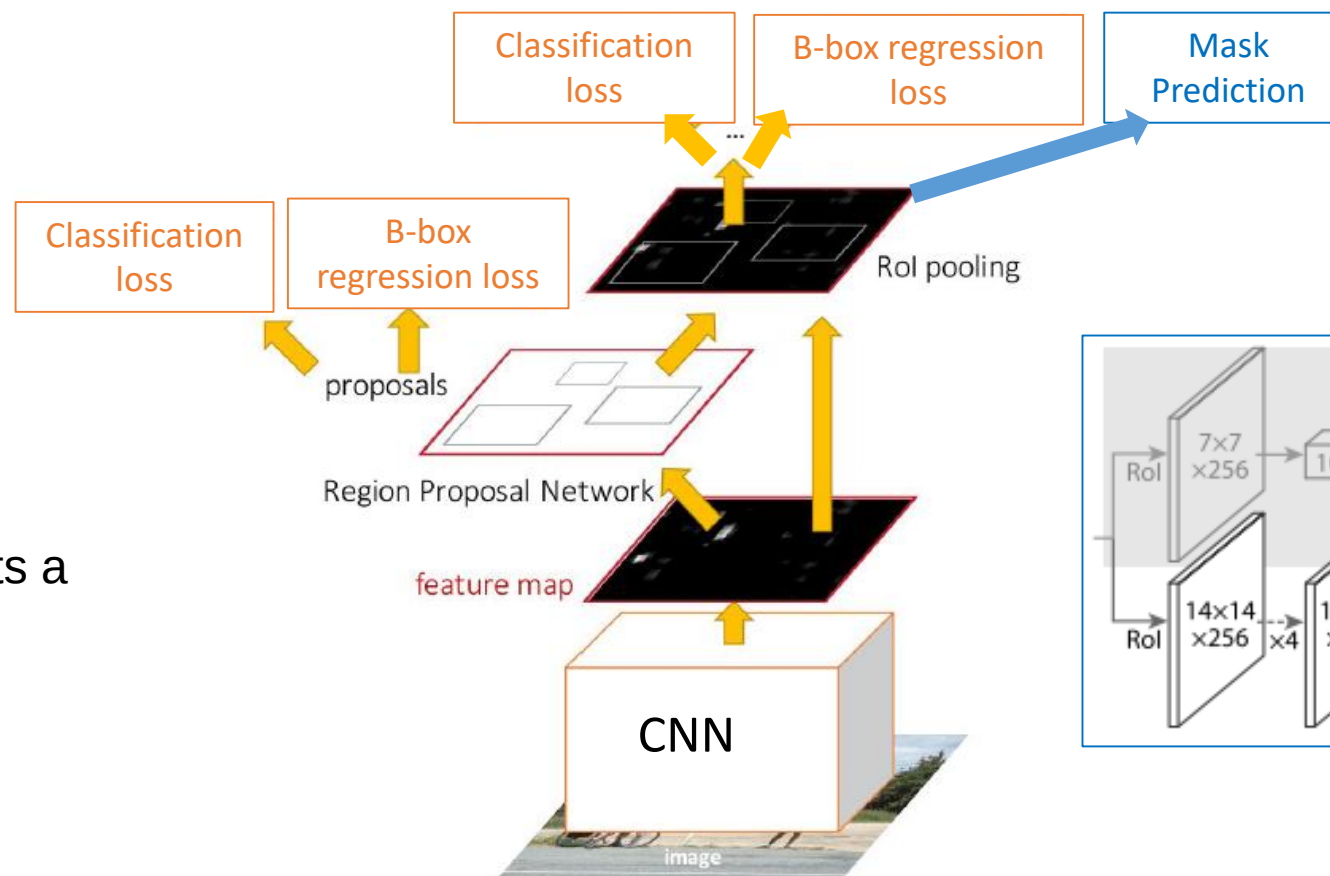
细胞图像中的实例分割

Instance Segmentation 实例分割

From Faster R-CNN to Mask R-CNN

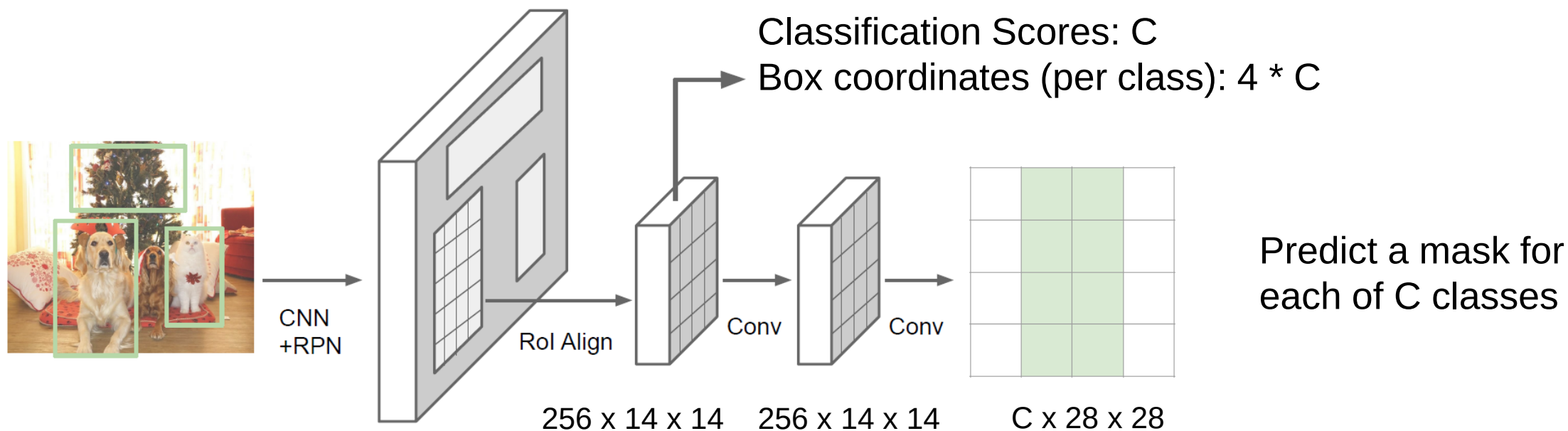
- 在Faster R-CNN得到的检测结果的基础上，使用一个额外的分支（如FCN）对ROI内进行分割

Add a small mask network that operates on each RoI and predicts a 28x28 binary mask



Instance Segmentation 实例分割

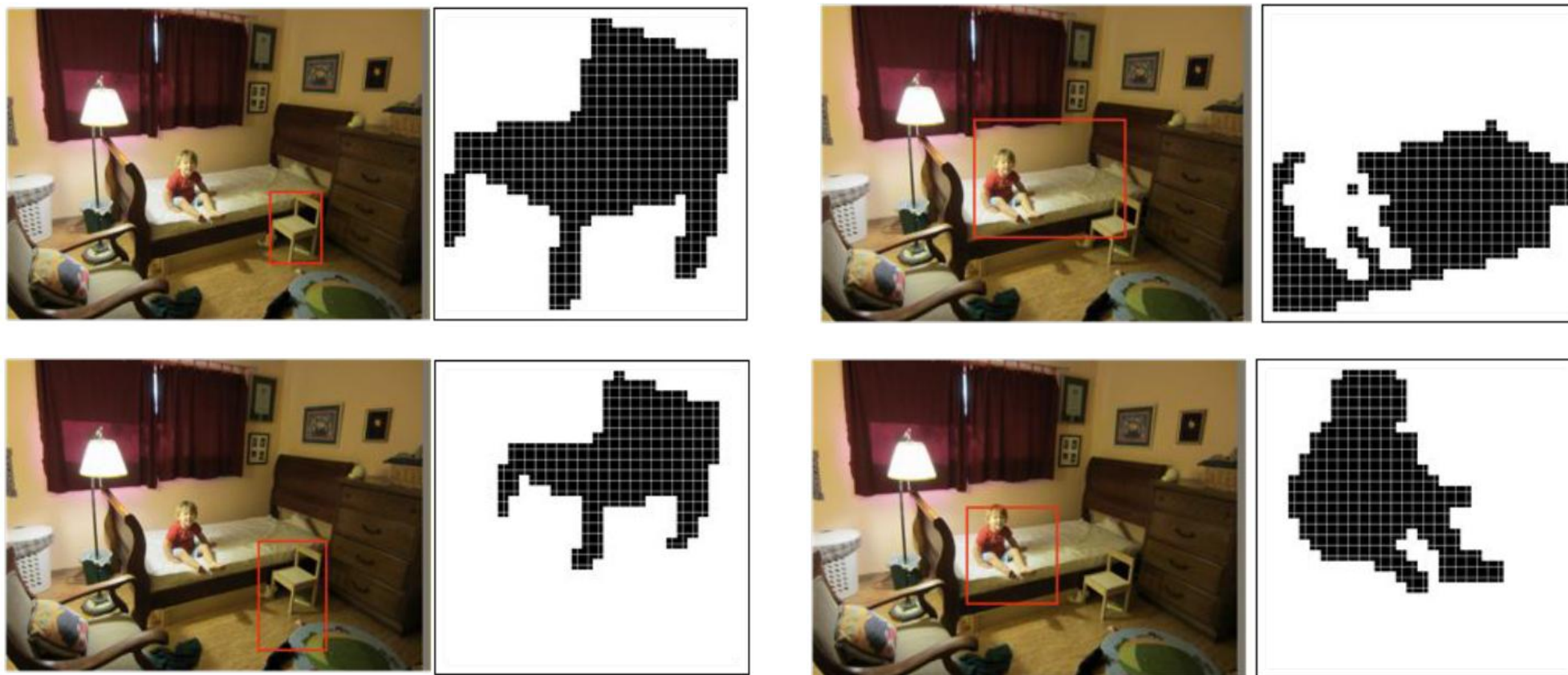
From Faster R-CNN to Mask R-CNN



Instance Segmentation 实例分割

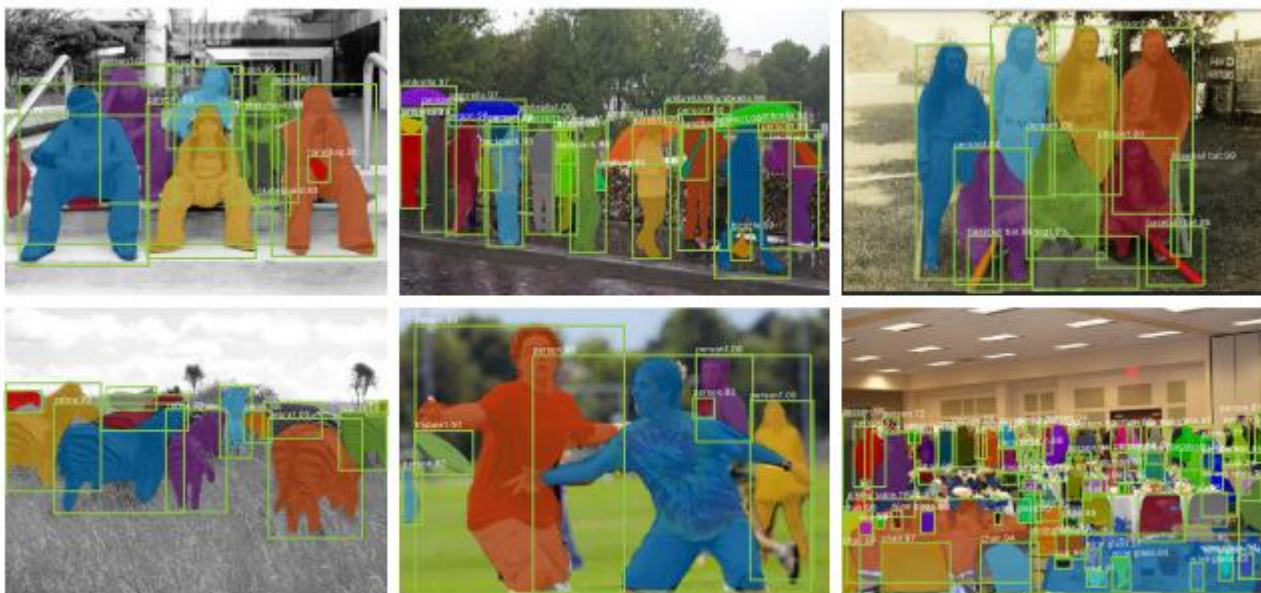
From Faster R-CNN to Mask R-CNN

检测结果与
分割结果



Instance Segmentation 实例分割

From Faster R-CNN to Mask R-CNN



Very Good Results !



Can also be used for pose estimation

Content



1, Image Segmentation

- FCN, U-Net, HRNet, Deeplab

2, Instance segmentation

- Mask-RCNN

3, Transfer learning

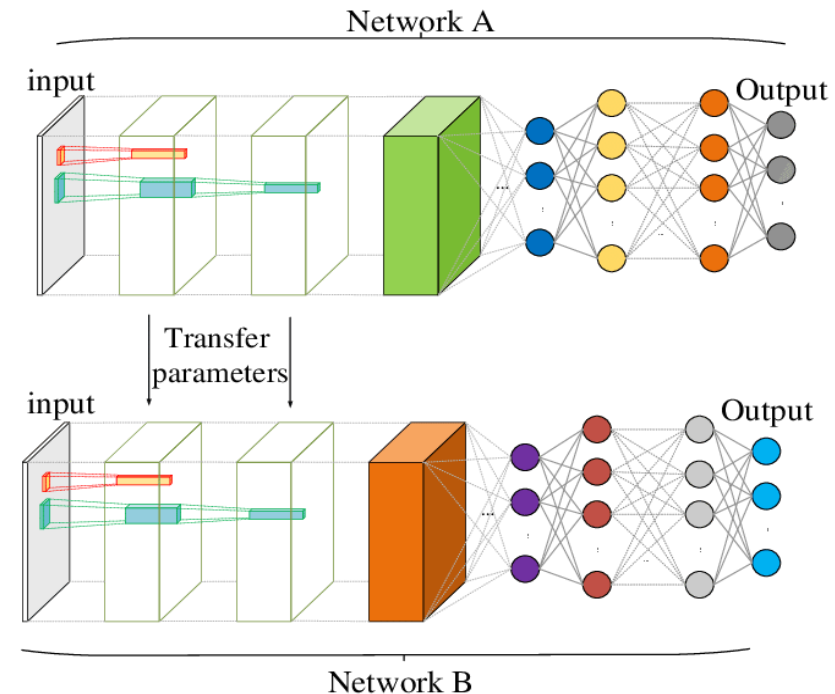
迁移学习/Transfer Learning

- The success on ImageNet relies on a large data set for training
- Usually we only have a small dataset available in a specific application
- Solution: **Transfer the knowledge from a large dataset** (ImageNet) to a given task

A **large** database



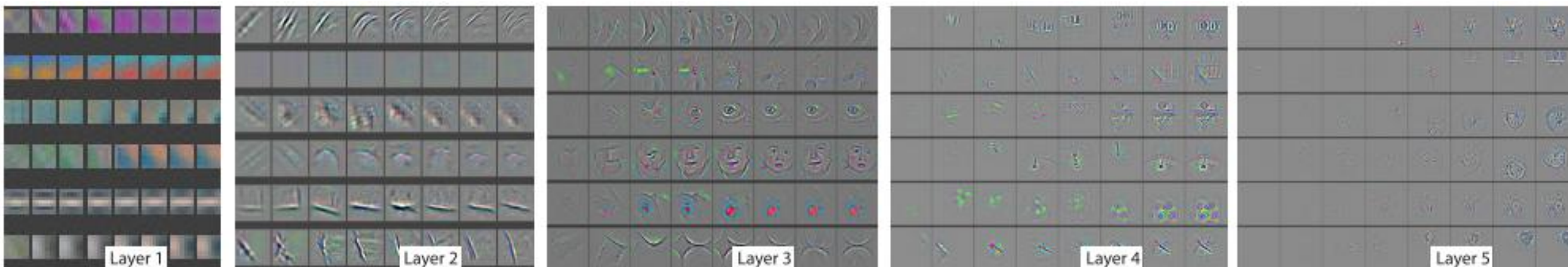
Smaller medical
image database



Why Transfer Learning

- The shallow layers learn general features that can be shared by different datasets

浅层学习到的是通用的特征，因此可以迁移到其他数据集或者任务上



Shallow layers learn some **general low-level patterns**, such as edge filters

Deep layers learn more **specific high-level patterns** associated with objects

Transfer Learning for Classification

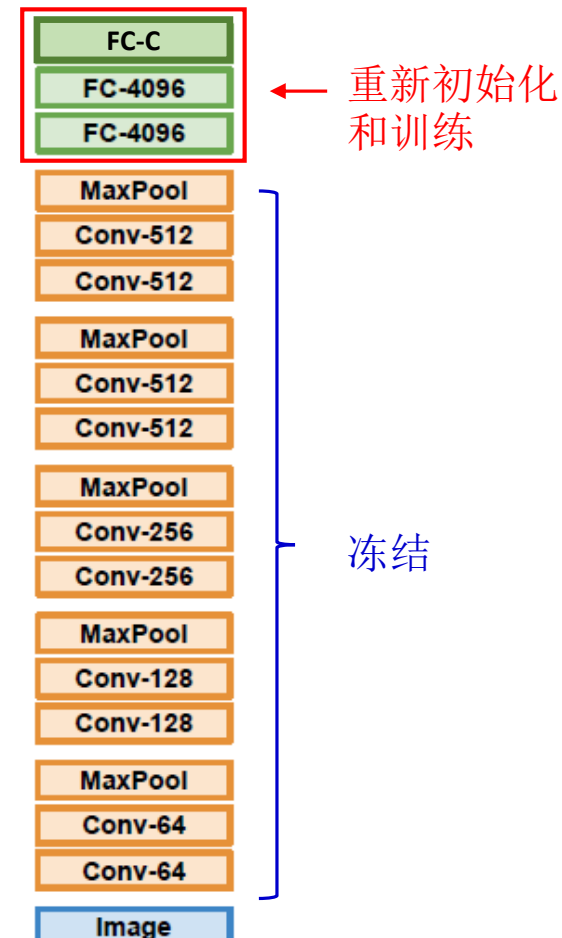
1. Train on ImageNet



2. Small Dataset (C classes)



3. Bigger Dataset (C classes)



Transfer Learning for Classification

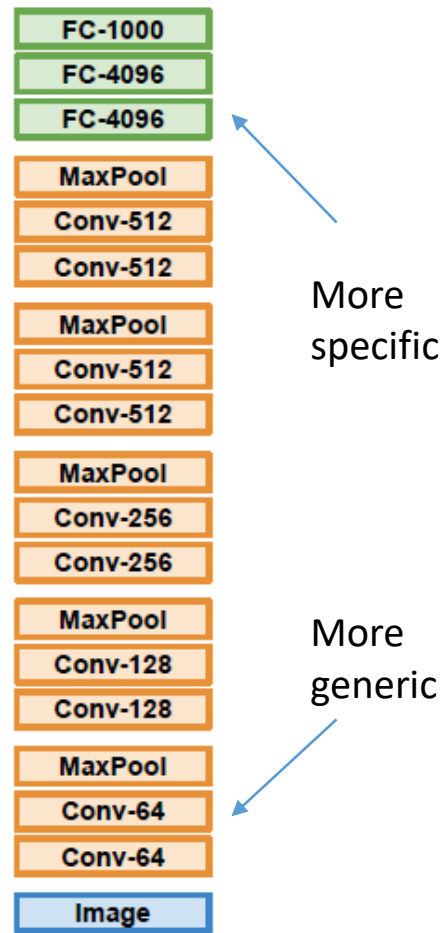


Hot to finetune（参数微调）？

Lower learning rate when finetuning; 1/10 of original LR is good starting point

	Very similar dataset	Very different dataset
Very little data	?	?
Quite a lot of data	?	?

Transfer Learning for Classification



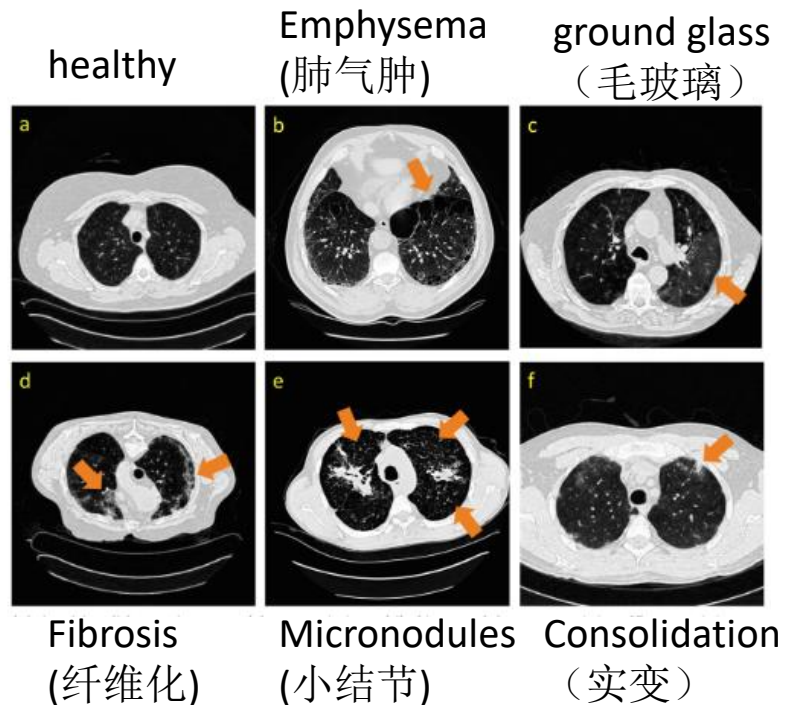
Hot to finetune (参数微调) ?

Lower learning rate when finetuning; 1/10 of original LR is good starting point

	Very similar dataset	Very different dataset
Very little data	Finetune linear classifier on top layer	You're in trouble... Try data augmentation / collect more data
Quite a lot of data	Finetune a few layers	Finetune a larger Number of layers

Transfer Learning Example

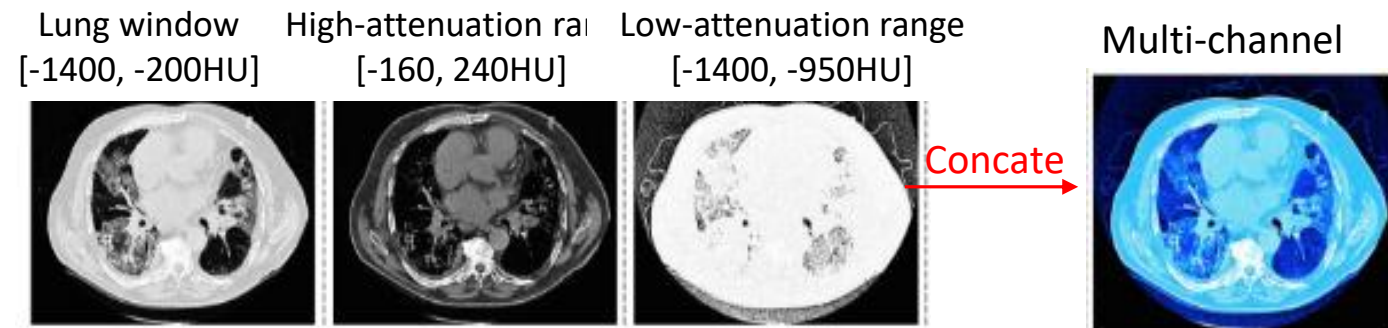
- Lung nodule classification



Interstitial Lung Disease Dataset

- Publicly available <https://www.nhlbi.nih.gov/health-topics/interstitial-lung-diseases>
- Contains 905 CT images from 120 patients
- 6 lung tissue types

Convert CT images to RGB, as Pre-trained models deal with RGB images

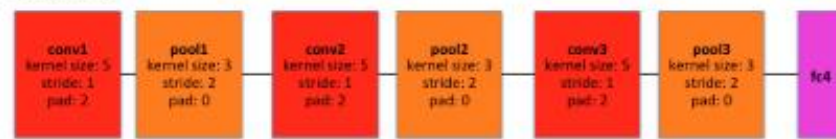


Transfer Learning Example

- Lung nodule classification

Different networks for finetuning (simplified illustration)

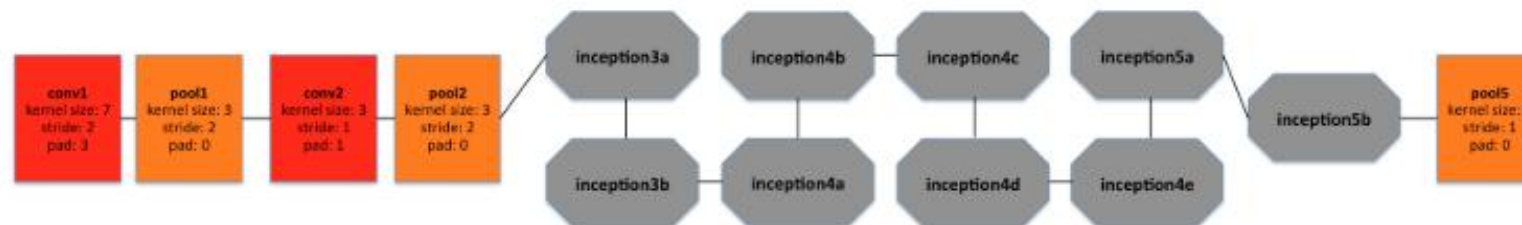
CifarNet



AlexNet



GoogleNet



仅微调最后的全连接层
(用较小的学习率)

Transfer Learning Example

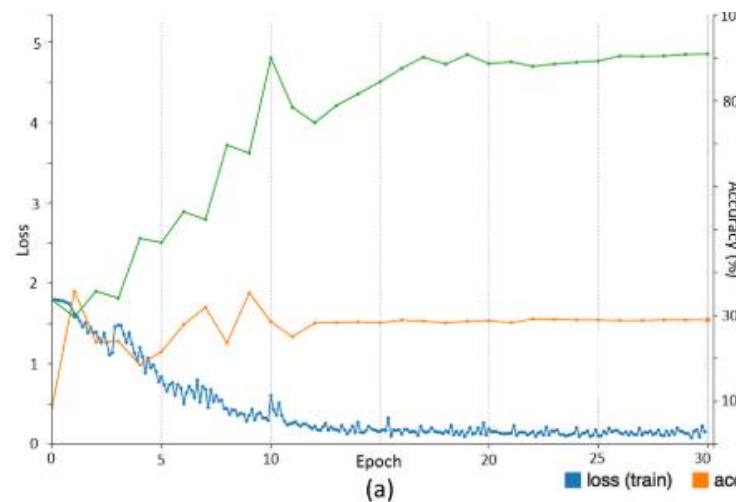


- Lung nodule classification

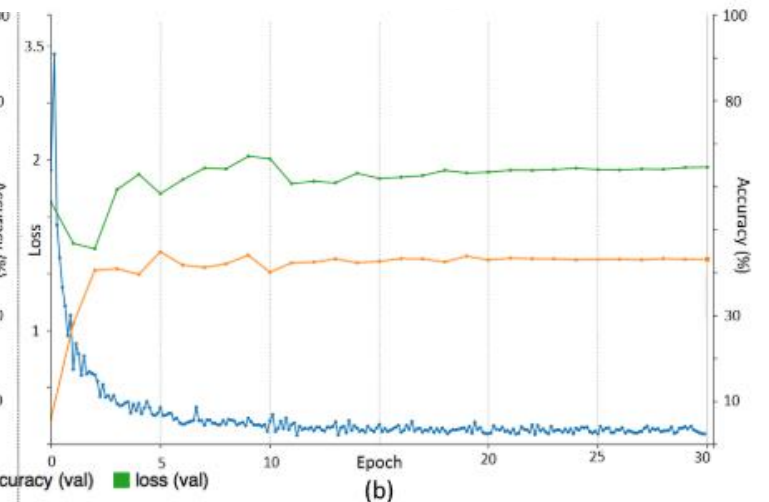
Classification accuracy

Method	ILD-Slice
CifarNet	-
AlexNet-TL	0.867
Overfeat-TL	0.877
VGG-16-TL	0.90
GoogLeNet-TL	0.902

AlexNet trained from random initialization



AlexNet fine-tuned from ImageNet CNN

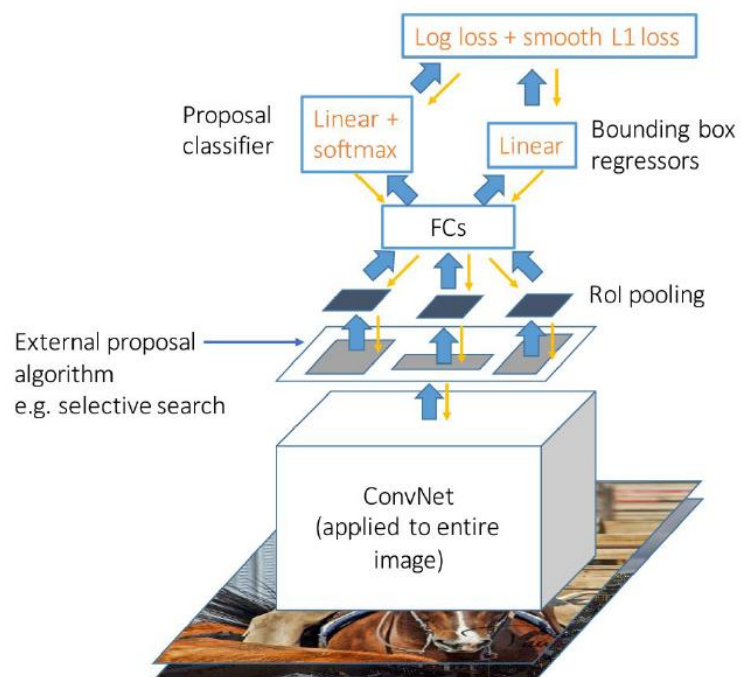


■ loss (train) ■ accuracy (val) ■ loss (val)

Transfer Learning for Other Tasks

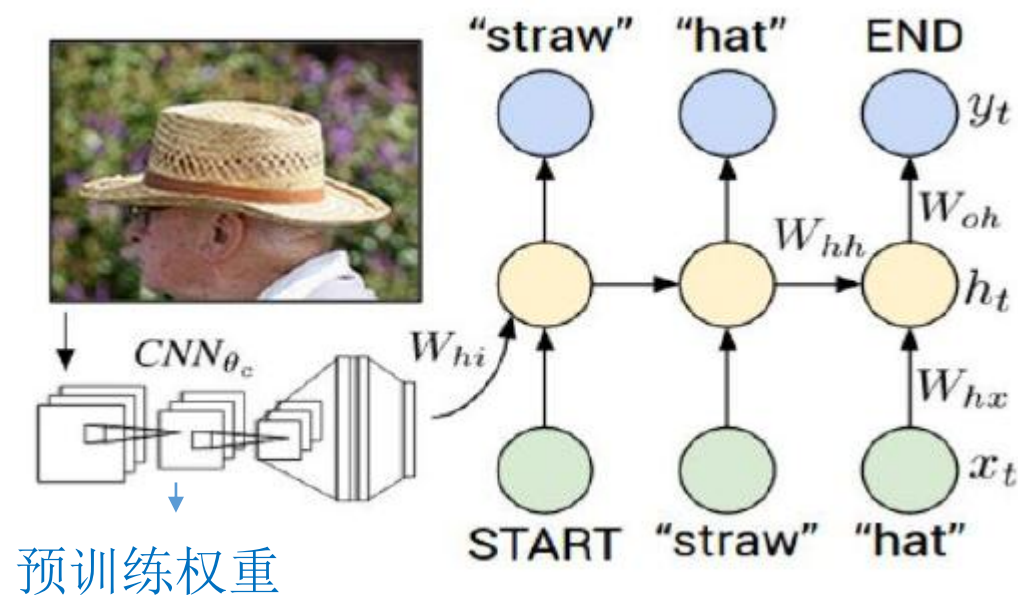
- Transfer learning is pervasive

Object Detection (Fast R-CNN)



Girshick, "Fast R-CNN", ICCV 2015

Image captioning



Karpathy and Fei-Fei, "Deep Visual-Semantic Alignments for Generating Image Descriptions", CVPR 2015

小结

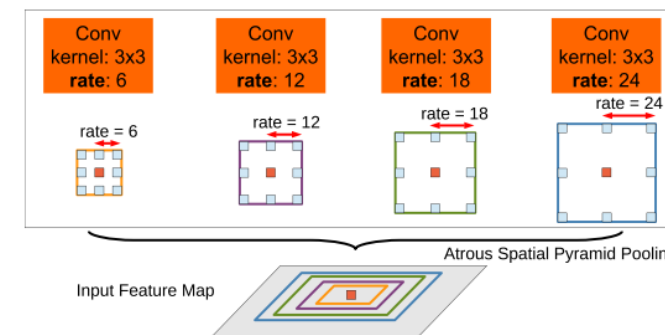
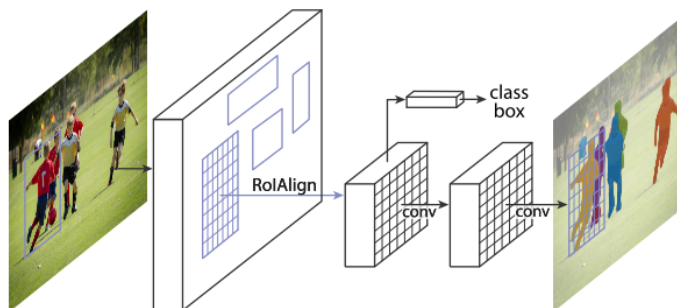
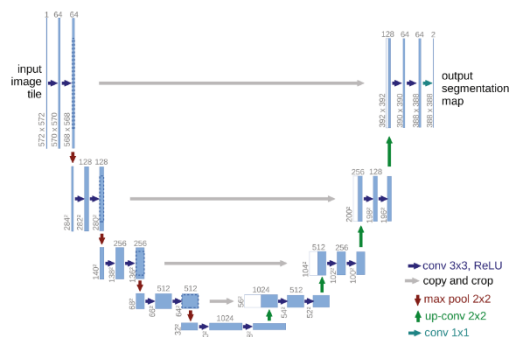
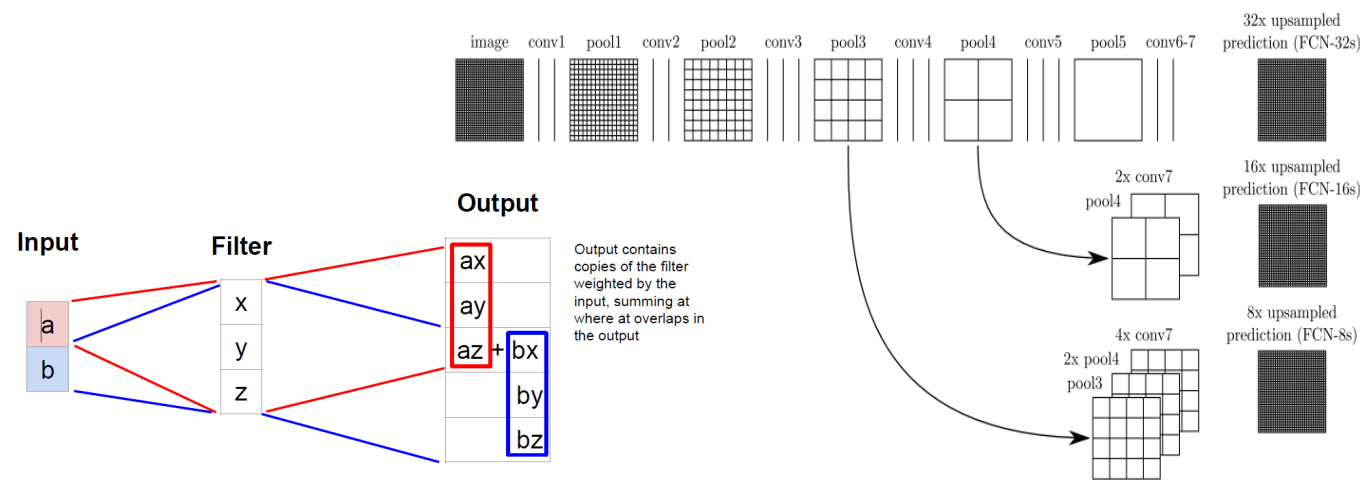
1, Image Segmentation

- FCN, U-Net, HRNet, Deeplab

2, Instance segmentation

- Mask-RCNN

3, Transfer Learning



小结



1. 关于反卷积说法正确的是?
 - A. 可用于特征图的下采样
 - B. 其输入通道和输出通道数量保持一致
 - C. 输入的步长小于输出的步长
 - D. 在FCN中用于图像分割任务

2. “Mask R-CNN既可以得到检测结果，又可以得到分割结果”是否正确?
 - A. 正确
 - B. 错误

3. “CNN中越深的层学到的特征更加通用”是否正确?
 - A. 正确
 - B. 错误